

Power BI Assignment 2

DAX and Data Visualization Report

Field	Details
Assignment Topic	DAX and Data Visualization
Tool Used	Power BI Desktop
Files Used	List of Orders.csv, Order Details.csv, Sales target.csv
Report Date	23 June 2026
Current Status	Re-checked and ready for final review/submission

1. Objective

The objective of this assignment was to use DAX calculations and Power BI visuals to analyze e-commerce sales data. The final report includes calculated columns, measures, and separate visualization pages for sales, targets, profit, location, sub-category distribution, and order count analysis.

2. Data Preparation Confirmation

- Assignment 1 Power BI file was duplicated and renamed for Assignment 2.
- Data was refreshed and the Sales Target cyclic reference issue was corrected.
- Sales Target columns were confirmed as Month of Order Date, Category, and Target.
- Amount, Target, and sales-related values were formatted as currency/fixed decimal values where required.
- City and State columns were created from the Location column for geographic and state-wise analysis.

3. Calculated Columns Created

Table	Calculated Column	Purpose / Formula Summary
Order Details	Category Type	Combined Category and Sub-Category to create a detailed product grouping.
Order Details	Revenue per Order	Calculated revenue using Amount multiplied by Quantity.
Order Details	Sales Category	Classified orders as Above Average or Below Average based on average Amount.
List of Orders	City	Extracted city name from Location before the comma.
List of Orders	State	Extracted state name from Location after the comma and removed extra spaces.

4. Measures Created

Measure	Home Table	Purpose
Order Count	Order Details	Counts distinct Order ID values.
Average Profit in Delhi	Order Details	Calculates average profit for orders related to Delhi.
YTD Sales	Order Details	Calculates year-to-date sales using Total Sales Amount and Order Date.
Total Sales Amount	Order Details	Helper measure used in visuals to summarize total sales amount.
Total Target	Sales Target	Helper measure used in target comparison visuals.

5. Data Visualization Pages Created

Page	Visual Type	Fields Used / Purpose
Sales Target Achievement by Category	Clustered column chart	Category on X-axis; Total Sales Amount and Total Target on Y-axis.
Max Profit Margin by Sub-Category	Donut chart	Sub-Category as legend/category; Max of Profit Margin as value.
Monthly Sales Trend	Line chart with Year slicer	Month on X-axis; Total Sales Amount on Y-axis; Year slicer for 2018 and 2019.
Profit and Quantity by Sub-Category	Scatter chart	Quantity on X-axis, Profit on Y-axis, and Sub-Category as legend.
Total Sales Amount and Target Cards	Cards and multi-row card	Cards for Total Sales Amount and Total Target; multi-row card for Minimum Target by Category.
Sales Performance Matrix	Matrix	Category and Month used to compare Total Target and Total Sales Amount.
Geographic Sales Analysis	Map	City as location and Total Sales Amount as bubble size.
Sales Distribution by Sub-Category	Treemap	Sub-Category as group and Total Sales Amount as value.
Order Count Analysis by State	Funnel chart	State as category and Order Count as value.

6. Re-check Summary

- The PBIX file contains 9 report pages, matching the required visualization list.
- The Monthly Sales Trend page includes a Year slicer, which helps view 2018 and 2019 separately.
- The Max Profit Margin donut chart uses Max of Profit Margin, which matches the assignment requirement.
- The Total Sales Amount and Target Cards page includes two cards, a page title, and a Minimum Target by Category multi-row card.
- The geographic map uses City and Total Sales Amount, which satisfies the geographic sales analysis requirement.
- The order count funnel uses State and Order Count, which satisfies the state-wise order count requirement.

7. Important Notes

- Some donut chart data labels may not show for every sub-category because Power BI hides crowded labels automatically. The legend and tooltip still show the information.
- The Sales Performance Matrix may require horizontal scrolling because it shows months and multiple values. This is acceptable for a matrix visual.
- The Monthly Sales Trend is best submitted with no year selected in the slicer, so the full month pattern is visible. The teacher can select 2018 or 2019 if needed.

8. Final Submission Checklist

- Save the Power BI file after final changes.
- Upload the PBIX file and this Word report to GitHub.
- Open the GitHub repository after upload and confirm both files are visible.
- Submit the GitHub repository link as the assignment submission.